WEEK 02 : TRIAL PRACTICE PHASE : DAYS 06 TO 10

Classical Machine Learning & Pipeline Engineering

Leak-Free Scikit-Learn Pipelines, Feature Engineering & TreeSHAP Explainability

PROGRAM & ROLE	CURRICULUM PHASE	TIME COMMITMENT	SCHEDULE & FEEDBACK
ML Engineering Intern (All-Round)	Trial Practice Tasks (Weeks 1-2)	15-20 Hours/Week (Part-Time Unpaid)	Submit by Friday (11:59 PM) | Weekend Feedback

01 Classical Machine Learning Philosophy & Practice Goals

PRACTICE PHASE

Building trustworthy machine learning models requires understanding that rigorous evaluation matters far more than algorithm complexity. In Week 2 of the Trial Practice Phase, you master leak-free scikit-learn pipeline design on real-world tabular practice datasets (such as customer churn, conversion, or risk classification). You will establish baseline lower bounds, engineer high-impact features, optimize gradient boosted trees, and audit predictions using TreeSHAP explainability before advancing to company product development in Week 3.

02 Leak-Free Pipeline Architecture & Baseline Benchmarks

DAYS 6-7

```
# Build leak-free preprocessing and model pipeline with ColumnTransformer
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.impute import SimpleImputer
# Preprocessing MUST be contained inside Pipeline so fits occur ONLY on training folds
preprocessor = ColumnTransformer(transformers=[
    ('num', Pipeline([('imputer', SimpleImputer(strategy='median')), ('scaler', StandardScaler())]), num_cols),
    ('cat', Pipeline([('imputer', SimpleImputer(strategy='constant', fill_value='UNKNOWN')), ('ohe', OneHotEncoder(handle_unknown='ignore'))]), cat_cols)
])
```

TASK 2.1 Baselines & Leak-Free Pipeline Architecture

Days 6-7 : Mon-Tue

Establish lower performance bounds with trivial baselines and benchmark 4 classical architectures inside a leak-free pipeline using stratified 5-fold cross-validation on the cleaned practice dataset.

- **Trivial Baselines:** Implement a Stratified Dummy Classifier and a Majority Class Predictor in `notebooks/03_baselines.ipynb` to define the empirical baseline performance floor.
- **Leak-Free Pipeline:** Encapsulate imputation, scaling, and encoding inside **`src/pipeline.py`** using `ColumnTransformer` and `Pipeline`. Preprocessing transformers must fit exclusively on training folds during cross-validation.
- **Model Benchmarking:** Train 4 distinct classical models: (1) Logistic Regression with L2 regularization, (2) k-Nearest Neighbors (k=7), (3) Decision Tree (max_depth=6), and (4) Random Forest (n_estimators=100, class_weight='balanced').
- **Validation Protocol:** Execute Stratified 5-Fold Cross-Validation on the training split. Touch the held-out test split strictly once. Report Accuracy, Precision, Recall, F1-Score, ROC-AUC, PR-AUC, and confusion matrix.

ACCEPTANCE CRITERIA

✓ All 4 classical models measurably beat the trivial baseline across PR-AUC and ROC-AUC metrics.
✓ Zero data leakage: mentor confirms scalers, encoders, and imputers fit solely on training folds.
✓ Random seeds explicitly pinned; cross-validation results are reproducible to within +-0.5% on rerun.

TASK 2.2 Feature Engineering & Hyperparameter Tuning

Days 8-9 : Wed-Thu

Engineer at least 5 meaningful domain features in **`src/features.py`**, perform an ablation study quantifying performance gains, and tune gradient boosting models to maximize classification performance.

- **5 Engineered Features:** Implement 5 domain features with documented rationale: (1) Numerical ratio/interaction features, (2) Categorical frequency/target encodings with out-of-fold regularization, (3) Cyclical time features (sine/cosine of timestamp hour/month), (4) Rolling or group aggregations, and (5) Non-linear binned variables.
- **Ablation Study:** Tabulate model performance with and without each feature group, recording ROC-AUC and PR-AUC deltas in `notebooks/04_tuning.ipynb`.
- **Hyperparameter Optimization:** Tune LightGBM and XGBoost via `RandomizedSearchCV` (or Optuna), optimizing learning rate, tree depth, subsample ratio, and regularization terms.
- **Artifact Serialization:** Export best-performing model to **`models/best_model.joblib`** with verified serialization integrity.

ACCEPTANCE CRITERIA

✓ Ablation table demonstrates clear, measurable ROC-AUC and PR-AUC performance lift from engineered features.
✓ Tuned gradient boosting model outperforms best baseline by a statistically significant margin with logged CV confidence intervals.
✓ Saved artifact **models/best_model.joblib** loads in an isolated Python session and executes inference correctly.

TASK 2.3 Model Explainability, Subgroup Auditing & Evaluation Report

Day 10 : Fri

Audit model explainability using TreeSHAP, inspect high-confidence error cases, verify subgroup fairness across user categories, and compile a comprehensive 3-4 page technical evaluation report.

- **TreeSHAP Interpretation:** Generate global SHAP beeswarm summary plots and mean absolute SHAP feature importance charts. Produce individual waterfall explanation plots for at least 3 individual predictions.
- **Error Analysis:** Isolate and inspect 20 high-confidence misclassifications (10 false positives, 10 false negatives). Categorize root causes and identify common failure patterns.
- **Subgroup Fairness Audit:** Evaluate performance parity across demographic or platform subgroups to verify absence of systemic model bias.
- **Technical Report:** Author a 3-4 page formal report in **`reports/week2_model_report.md`** documenting recommended architecture, validation metrics, error analyses, and Week 3 product transition plan.

ACCEPTANCE CRITERIA

✓ Report contains concrete operational recommendations, explicit probability decision thresholds, and known edge-case failure modes.
✓ Error analysis includes at least 3 concrete, evidence-backed engineering hypotheses for future model refinement.
✓ All interpretability assertions are supported by exported SHAP figures in **reports/figures/**.

ATTENOMY ENGINEERING STANDARD : PROBABILITY CALIBRATION

Tree-based models frequently output uncalibrated probability scores near 0.0 and 1.0. Prior to setting operational decision thresholds, evaluate probability calibration curves and apply **CalibratedClassifierCV(method='isotonic', cv='prefit')** where required.

03 Week 2 Deliverables & Artifacts Matrix

VERIFICATION MATRIX

ARTIFACT NAME	REPOSITORY TARGET PATH	PRIMARY REQUIREMENT	EVALUATION
Baseline Notebook	`notebooks/03_baselines.ipynb`	Trivial baselines + 4 classical models	Score: 18%
Pipeline Module	`src/pipeline.py`	Modular leak-free ColumnTransformer pipeline	Score: 15%
Feature Engineering	`src/features.py`	5 typed features with unit tests	Score: 18%
Tuning Notebook	`notebooks/04_tuning.ipynb`	Ablation study + LightGBM/XGBoost tuning	Score: 18%
Best Model Artifact	`models/best_model.joblib`	Serialized pipeline, reloads cleanly	Pass / Fail
SHAP Figures	`reports/figures/shap_summary.png`	Beeswarm summary + local waterfall plots	Score: 15%
Evaluation Report	`reports/week2_model_report.md`	3-4 page formal report with error analysis	Score: 18%

04 Weekly Submission Deadline & Weekend 1-on-1 Feedback

GATE 2 ASSESSMENT

Weekly Submission Deadline: There is no fixed daily stand-up. Interns work flexibly (15-20 hours/week) and must submit all deliverables, pipeline code, model evaluation reports, and SHAP analyses to their practice repository by end of Friday (11:59 PM). On Saturday or Sunday, you will be invited to an online 1-on-1 meeting for personalized feedback and code defense. Passing Gate 2 is required to unlock Week 3 product development.

GATE 2 PASSING CONDITION & MENTOR PROBE QUESTIONS

✓ **Mandatory Pass Condition:** Leakage-free pipeline verified via code inspection, tuned model beating baseline by documented margin with CV std dev < 0.5%, and evaluation report delivered.

- **Technical Probe 1:** "Show me the exact line where test data is isolated during cross-validation. How do you guarantee the imputer median was not computed across the full dataset?"
- **Technical Probe 2:** "Which feature exhibited the strongest SHAP importance in your model, and how did you confirm it was not a spurious correlation?"
- **Technical Probe 3:** "How did your ablation study isolate the individual performance contribution of each engineered feature group?"

05 Actionable Daily Execution Checklist

INTERN CHECKLIST

- ☐ **Day 6:** Implement dummy baseline, design ColumnTransformer pipeline in src/pipeline.py.
- ☐ **Day 7:** Benchmark 4 classical models with stratified 5-fold CV, record metric table.
- ☐ **Day 8:** Implement 5 domain features in src/features.py, run ablation study in notebook.
- ☐ **Day 9:** Tune LightGBM/XGBoost hyperparameters, export best model artifact to joblib.
- ☐ **Day 10:** Generate SHAP plots, audit 20 errors, complete report, submit by end of Friday (11:59 PM), attend weekend review.
- ☐ **Reflect:** Log 3 technical entries in LEARNING_LOG.md with concepts and code links.

06 Flexible Engineering Cadence & Mentorship Support SLA

ENGINEERING CADENCE

Flexible Workflow: No fixed daily stand-ups. Interns manage their hours flexibly to meet the 15-20 hours/week commitment while pushing regular atomic git commits (`feat:`, `fix:`, `test:`).

Weekend Feedback: Every Saturday or Sunday, you will receive personalized 1-on-1 feedback and pipeline code reviews from an Attenomy ML Lead.

45-Minute Help Escalation Rule: When blocked by a bug, investigate independently for 45-60 minutes before messaging your mentor with: (1) your objective, (2) the error traceback, and (3) your troubleshooting attempts.